

SAN FRANCISCO BAY RESERVE, CA, USA

WMO: 998011

Lat: 38.2004N

Lon: 122.0260W

Elev: 10

StdP: 14.69

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	33.0	35.4	14.9	11.7	58.8	19.7	14.8	56.2	22.8	54.2	20.1	55.1	2.4	50	0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	24.9	92.8	66.0	90.2	65.2	86.6	64.3	68.5	87.1	67.3	84.6	66.0	82.5	7.8	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.8	82.6	73.5	60.4	78.4	72.4	59.4	75.7	71.5	32.6	87.1	31.6	84.6	30.6	82.3	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 20.1	18.4	16.9	DB	29.2	96.8	3.3	4.6	26.8	100.2	24.9	102.9	23.0	105.5	20.6	108.9	(4)
(5)			WB	27.5	71.0	3.0	1.3	25.3	71.9	23.6	72.7	21.9	73.4	19.7	74.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.0	48.8	51.4	55.1	58.9	62.4	67.2	69.3	69.6	69.7	64.2	54.9	48.1		
	DBStd	9.44	4.64	4.84	5.15	6.13	5.70	5.40	4.86	4.85	5.17	5.46	5.23	5.08		
	HDD50	243	76	38	13	2	0	0	0	0	0	0	16	98		
	HDD65	2495	501	380	308	204	121	33	12	11	13	81	305	526		
	CDD50	3897	40	78	171	268	385	517	597	608	592	441	162	38		
	CDD65	673	0	0	1	20	41	100	143	155	155	57	1	0		
	CDH74	6811	2	7	31	281	505	1119	1338	1366	1524	613	25	0		
	CDH80	2654	0	0	1	83	180	443	524	535	686	201	1	0		
(14) Wind	WSAvg	8.1	5.0	6.3	6.9	8.6	10.4	11.0	12.2	11.3	8.7	6.4	5.2	5.2		
(15) Precipitation	PrecAvg	21.2	4.5	4.1	2.4	1.4	0.9	0.2	0.0	0.0	0.0	1.1	2.1	4.4		
	PrecMax	34.4	11.5	12.6	9.6	4.6	3.1	1.2	0.0	0.6	0.2	4.5	6.5	11.0		
	PrecMin	11.1	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4		
	PrecStd	6.3	3.2	2.8	2.5	1.1	0.9	0.3	0.0	0.1	0.1	1.1	1.7	3.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.9	74.6	77.9	87.3	91.2	96.0	95.5	94.8	96.5	91.3	77.5	65.5		
		MCWB	50.9	55.7	60.1	63.3	64.8	65.7	67.9	68.6	63.8	62.0	59.3	51.6		
	2%	DB	65.1	68.6	73.1	81.9	85.2	90.6	90.7	91.0	92.0	85.7	73.5	61.8		
		MCWB	51.6	55.2	58.0	61.8	60.9	65.6	67.2	66.8	64.6	61.3	57.9	53.2		
	5%	DB	61.4	64.3	69.6	76.1	79.3	86.0	86.1	86.3	88.9	80.5	69.2	59.3		
		MCWB	51.5	53.3	55.8	59.8	60.4	64.4	66.2	65.7	63.7	59.3	55.3	52.1		
	10%	DB	58.3	61.0	65.9	70.9	74.2	80.9	81.3	81.8	83.3	76.4	64.9	56.9		
		MCWB	50.9	52.1	54.2	57.3	59.0	62.5	65.1	65.0	62.9	59.1	54.0	51.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.6	60.2	62.0	66.5	66.6	69.9	70.9	70.7	68.0	66.5	62.6	59.7		
		MCDB	61.8	68.0	74.9	83.7	87.6	90.5	92.3	90.8	88.1	80.4	73.5	61.8		
	2%	WB	55.4	57.9	59.4	63.4	63.8	67.8	68.3	68.3	66.4	64.2	60.0	57.4		
		MCDB	60.1	64.3	69.8	78.4	80.9	86.0	86.4	85.9	84.6	78.1	69.6	58.7		
	5%	WB	53.5	56.0	57.6	60.8	61.7	65.8	66.7	66.6	65.3	62.9	58.1	55.4		
		MCDB	58.3	61.8	66.0	73.7	76.6	82.3	83.4	82.9	82.5	75.3	65.2	56.9		
	10%	WB	51.9	54.0	55.9	58.7	59.7	63.8	65.3	65.3	64.1	61.4	56.4	53.5		
		MCDB	56.1	59.4	63.2	69.5	72.4	78.5	80.7	80.4	80.2	72.0	62.4	55.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.7	17.5	18.5	20.5	20.6	23.1	23.8	24.3	24.9	22.4	19.1	15.4		
		MCDBR	24.1	24.2	24.8	28.5	28.0	30.8	30.0	30.0	32.3	30.3	25.5	18.9		
		MCWBR	12.8	12.9	11.9	12.3	11.4	11.6	11.2	10.4	10.1	11.4	12.0	10.8		
	5% WB	MCDBR	18.3	19.6	20.8	26.1	25.0	27.3	27.4	26.7	27.2	23.8	20.7	11.4		
		MCWBR	11.8	11.7	10.7	12.1	11.5	11.8	11.2	10.5	9.9	9.8	11.0	8.2		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.323	0.337	0.357	0.361	0.359	0.362	0.348	0.346	0.328	0.323	0.325		
	taud		2.531	2.530	2.478	2.398	2.411	2.421	2.410	2.442	2.451	2.534	2.528	2.521		
	Ebn at Noon		274	289	294	292	291	291	289	292	286	283	270	263		
	Edh at Noon		24	28	32	37	37	36	37	35	32	27	24	23		
(44) All-Sky Solar Radiation	RadAvg		722	1036	1470	1938	2309	2559	2506	2237	1844	1319	874	638		
	RadStd		119	132	146	162	139	105	73	72	63	87	60	102		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)	+0.68	N/A	-2.02	N/A	N/A	N/A	N/A	-211	+236	N/A

Nomenclature: See separate page